NAVRACHANA UNIVERSITY

Teaching Plan

Mobile Application Development (CS344)

Credit : 2

Faculty	Name	Contact	Office Hours	Office Location
	Dharmeshkumar Rana	dharmeshkumar.rana@nuv.ac.in		

School	SET

Semester	Autumn (July to Dec 2025)
Lecture time & Weedays	SECTION_1 | Saturday | 14:00:00-16:00:00 | 26-07-2025 TO 14-11-2025 SECTION_2 | Saturday | 11:00:00-13:00:00 | 26-07-2025 TO 14-11-2025

Pre-requisite	

Co-requisite	

Anti-requisite	
Course Description	This course was designed to Mobile computing devices have become ubiquitous in our communities. In this course, we focus on the creation of mobile solutions for various modern platforms, including major mobile operating systems. Topics include mobile device architecture, programming languages, software engineering, user interface design, and app distribution.
Course Abstract	Mobile application development course focus on process of creating software applications that run on a mobile device, and a typical mobile application utilizes a network connection to work with remote computing resources. Hence, the mobile development process involves creating installable software bundles (code, binaries, assets, etc.), implementing backend services such as data access with an API, and testing the application on target devices.
Course Objectives	1. Design mobile based applications. 2. Able hybrid application development platform. 3. Develop mobile based applications. 4. Analyze and troubleshoot the problems while developing application. 5. Deploy application on available online app stores.
Course Outcomes	Students will be able to: 1) Understand the basic concepts of Mobile Application development and its type 2) Understand the basic android application development and its deployment process. 3) Implement responsive Android Applications with various component, activities, layout managers and event handlers. 4) Make the use of data storage like content providers, shared preferences and SQLite database. 5) Present the concepts of iOS app development as part of case study
Course Typology	Theory
Course Outline	Unit- I: Introduction to Mobile Application Development Introduction (4 Hours) Importance of Mobile Applications, Types of Mobile Applications: Native, Hybrid and Web Applications, advantages and disadvantages of Different types of Applications, Application Development Environment, Smartphone Application Platforms: Android, iOS and Windows, Similarities and differences between platforms, Mobile Application development process – Mobile Software Engineering, Frameworks and Tools. Unit- II: Introduction to Android (6 Hours) Introduction to Android Platform, History of Android, Virtual device, development tools, Java packages, emulators, services, Structure and lifecycle of an application for Android system, Android Layouts, Deployment of applications:localization of applications, application signing, version management, licenses. Unit- III: Android XML and GUI based Design (6 Hours) Graphical User Interface: preparing containers and components, management of component layout, event handling, XML functions and keywords, Responsive designs. Unit- IV: Content Providers and Database (8 Hours) Processing of application resources, content providers, file system, Data persistence: backups, databases, Application security and permissions: security architecture, application signing, user identification, file access, declaration and verification of permissions, Network communication and internet applications, Wi-Fi connections, Geographical location: use of GPS data, Bluetooth communication: basics, permissions, Bluetooth device discovery, device connectivity as a client, server creation, connection management, Push notification, Database design and implementation: Interfacing with SQLite, use of preferences and content providers. Unit- V: Introduction to iOS and Xcode (Case Study) (6 Hours) IOS Architecture, Frameworks, MVC Design Pattern, Application Life Cycle, Features, Introduction to Objective-C, General Concepts of Objective-C, Introduction to Swift, General Concepts of Swift, Introduction to Xcode, Deployment in App store.
Pedagogy	This course will be based on Project based learning, in which students will be assigned course project in which they need to apply the knowledge gained during the theoretical and practical sessions. Also for Unit-5, there will be flip class-room method will be applied where students will give presentation on allotted topics.
Expectations from Students	1) Maintain silence in class. 2) Students should attend 100% classes. 3) Don’t distract or harass others 4) Participate in class discussions 5) Turn in and do well in Assignments/Homework 6) Demonstrate in class presentation/discussions 7) Pass in Midterm and Final exam 8) Arrive to class on time 9) Maintain 80% attendance (minimum) 10) No plagiarism at all 11) Come prepared to class
Assessment / Evaluation	
Project / Assignment Details	Will be discussed in class.
Course Material	-
Additional Information	-

Session Plan

Topic Title	Session No.	Topic & Subtopic Details	Readings, Cases, etc.	Activities	Important Dates
Introduction, Importance of Mobile Applications	1	Introduction, Importance of Mobile Applications		Task/Project	
Types of Mobile Applications: Native, Hybrid and Web Applications, advantages and disadvantages of Different types of Applications	2	Types of Mobile Applications: Native, Hybrid and Web Applications, advantages and disadvantages of Different types of Applications		Task/Project	
Application Development Environment, Smartphone Application Platforms: Android, iOS	3	Application Development Environment, Smartphone Application Platforms: Android, iOS		Task/Project	
Similarities and differences between platforms	4	Similarities and differences between platforms		Task/Project	
Introduction to Android Platform, History of Android, Virtual device,	5	Introduction to Android Platform, History of Android, Virtual device,		Task/Project	
development tools, Java packages, emulators,	6	development tools, Java packages, emulators,		Task/Project	
services, Structure and lifecycle of an application for Android system,	7	services, Structure and lifecycle of an application for Android system,		Task/Project	
Android Layouts	8	Android Layouts		Task/Project	
Android Layouts	9	Android Layouts		Task/Project	
Android Layouts	10	Android Layouts		Task/Project	
Deployment of applications: localization of applications,	11	Deployment of applications: localization of applications,		Task/Project	
application signing, version management, licenses.	12	application signing, version management, licenses.		Task/Project	
Android XML and GUI based Design Graphical User Interface:	13	Android XML and GUI based Design Graphical User Interface:		Task/Project	
preparing containers and components	14	preparing containers and components		Task/Project	
management of component layout	15	management of component layout		Task/Project	
event handling	16	event handling		Task/Project	
XML functions and keywords, Responsive designs.	17	XML functions and keywords, Responsive designs.		Task/Project	
Processing of application resources, content providers,	18	Processing of application resources, content providers,		Task/Project	
file system, Data persistence:	19	file system, Data persistence:		Task/Project	
Application security and permissions: security architecture,	20	Application security and permissions: security architecture,		Task/Project	
file access, declaration and verification of permissions	21	file access, declaration and verification of permissions		Task/Project	
Network communication and internet applications, Wi-Fi connections,	22	Network communication and internet applications, Wi-Fi connections,		Task/Project	
Geographical location: use of GPS data,	23	Geographical location: use of GPS data,		Task/Project	
Bluetooth communication: basics, permissions, Bluetooth device discovery	24	Bluetooth communication: basics, permissions, Bluetooth device discovery		Task/Project	
device connectivity as a client, server	25	device connectivity as a client, server		Task/Project	
creation, connection management, Push notification,	26	creation, connection management, Push notification,		Task/Project	
Introduction to iOS and Xcode (Case Study) IOS Architecture, Frameworks, MVC, Design Pattern, Application Life Cycle	27	Introduction to iOS and Xcode (Case Study) IOS Architecture, Frameworks, MVC, Design Pattern, Application Life Cycle		Task/Project	
Introduction to Objective-C,General Concepts of Objective-C	28	Introduction to Objective-C,General Concepts of Objective-C		Task/Project	
Introduction to Swift, General Concepts of Swift,	29	Introduction to Swift, General Concepts of Swift,		Task/Project	
	30	IOT Project			

Publish Status	Published